Polymer descriptors from SMILES

Features used to predict Tg, FFV, thermal conductivity (Tc), density and radius of gyration (Rg) with machine learning

All descriptors are calculated with RDKit from the SMILES of the **repeat unit**, in which two ***** atoms mark where the unit connects to its neighbours. The **backbone** is the shortest path between the two * atoms; everything else is **side chain**. Example: in PET the ester sits in the backbone, in PMMA in the side chain.

Group	Count	What it describes
A · Original	9	Size, polarity, rings, H-bonding
B · Chemistry	12	Flexibility, composition, backbone vs. side chain
B+ · Functional groups	22	Ester, amide, imide … in backbone or side chain
Chain (new)	23	Backbone flexibility, ring linkages, side chains, volume
Shape (new)	7	3D shape of three joined repeat units
D · Morgan fingerprints	1,024 bits	Local atom environments
Total	**73 + 1,024 bits**	

A · Original descriptors (9)

Descriptor	Meaning
MolWt	Molar mass of the repeat unit
HeavyAtomCount	Number of non-hydrogen atoms
NumAtoms	Number of atoms (incl. the two *)
NumRings	Number of rings
NumAromaticRings	Number of aromatic rings
NumHDonors	Hydrogen-bond donors (OH, NH)
NumHAcceptors	Hydrogen-bond acceptors (O, N)
MolLogP	Hydrophobicity (octanol/water partition coefficient)
TPSA	Topological polar surface area

B · Chemistry descriptors (12)

Descriptor	Meaning
NumRotatableBonds	Flexible bonds in the whole repeat unit
FractionCSP3	Share of sp^3 carbons (aliphatic vs. aromatic)
AromaticAtomFraction	Share of heavy atoms that are aromatic
HalogenCount	F, Cl, Br and I atoms
SulfurCount	S atoms
SiliconCount	Si atoms (siloxanes)
MolWtPerHeavyAtom	Average mass per heavy atom
MolLogPPerHeavyAtom	Hydrophobicity per heavy atom
TPSAPerHeavyAtom	Polar surface per heavy atom
BackboneLength	Atoms in the backbone per repeat unit
BackboneRingFraction	Share of backbone atoms that are in a ring
SideChainFraction	Share of heavy atoms outside the backbone

B+ · Functional-group descriptors (22)

*Groups are counted on three joined repeat units, so groups that cross the * atoms (e.g. the carbonate of polycarbonate) are found, and each group is assigned to the backbone or a side chain.*

Descriptor	Meaning
EsterBackbone / EsterSideChain	Esters, e.g. PET (backbone) vs. PMMA (side chain)
CarbonateBackbone / …SideChain	Carbonates, e.g. polycarbonate
AmideBackbone / …SideChain	Amides, e.g. nylons and aramids
ImideBackbone / …SideChain	Imides, e.g. polyimides
UrethaneBackbone / …SideChain	Urethanes
EtherBackbone / …SideChain	Ethers, e.g. PEO, PEEK
SiloxaneBackbone / …SideChain	Si–O–Si, e.g. PDMS
SulfoneBackbone / …SideChain	Sulfones, e.g. polysulfone
HydroxylSideChain	OH groups
NitrileSideChain	C≡N groups, e.g. polyacrylonitrile
LargestSideGroup	Heavy atoms in the largest side group
BackboneUnsaturation	Share of non-aromatic double/triple bonds in the backbone
HDonorsPerHeavyAtom	H-bond donors per heavy atom
HAcceptorsPerHeavyAtom	H-bond acceptors per heavy atom

Chain descriptors (23) · new

A bond counts as flexible if it is a single bond, not in a ring, and not an amide C–N bond (partial double-bond character).

Descriptor	Meaning
Backbone flexibility	
BackboneRotatableBonds	Flexible bonds in the backbone
BackboneRotatableFraction	Share of backbone bonds that are flexible
LongestFlexibleRun	Longest run of consecutive flexible backbone bonds (spacer length, e.g. $–CH_2–CH_2–CH_2–$)
FullyFlexibleBackbone	1 if every backbone bond is flexible (e.g. PE, PS, PDMS)
BackboneAromaticFraction	Share of backbone atoms that are aromatic
KierFlexibility	Kier flexibility index Φ
Ring linkages	
BackboneRingCount	Rings the backbone passes through
ParaLinkages	Para-linked benzene rings in the backbone (straight, stiff; e.g. Kevlar)
MetaLinkages	Meta-linked benzene rings in the backbone (kinked; e.g. Nomex)
OrthoLinkages	Ortho-linked benzene rings in the backbone
ParaLinkageFraction	Share of backbone benzene rings that are para-linked
Side chains	
NumSubstituents	Groups attached to the backbone (incl. single atoms such as =O, F, CH_3)
NumSideChains	Substituents with at least two heavy atoms
LongestSideChain	Atoms in the longest substituent, counted from the backbone outwards
MeanSubstituentSize	Average number of heavy atoms per substituent
SideChainRotatableBonds	Flexible bonds in the substituents (internal plasticisation)
SideChainBranchPoints	Substituent atoms with three or more heavy neighbours (branching, bulk)
PendantAromaticRings	Aromatic rings in substituents (e.g. the phenyl of polystyrene)
QuaternaryCarbons	Carbons with four carbon neighbours (e.g. tert-butyl, bisphenol A)
Volume and packing	
VdWVolume	Van der Waals volume of the repeat unit, $Å^3$ (Zhao et al. 2003)
SpecificVdWVolume	Van der Waals volume per mass, cm^3/g (FFV = $(V - 1.3 \cdot V_w)/V$)
VdWVolumePerHeavyAtom	Van der Waals volume per heavy atom
LabuteASAPerVolume	Surface area per volume (compact vs. extended groups)

Shape descriptors (7) · new

Three repeat units are joined, the ends capped with hydrogen, and one 3D conformer is generated (RDKit ETKDG). One conformer is only a rough picture of a real chain.

Descriptor	Meaning
TrimerRadiusOfGyration	Radius of gyration of the trimer, Å
TrimerRgPerBackboneAtom	Radius of gyration per backbone atom
TrimerEndToEndRatio	End-to-end distance relative to a fully stretched chain
TrimerAsphericity	Deviation from a sphere (0 = sphere, 1 = rod)
TrimerEccentricity	Elongation of the shape
TrimerNPR1, TrimerNPR2	Normalised principal moments of inertia (rod, disc or sphere)

D · Morgan fingerprints (1,024 bits)

Descriptor	Meaning
Morgan_0 … Morgan_1023	Each bit marks whether a particular atom environment (an atom and its neighbours up to two bonds away) occurs in the repeat unit

Not available from SMILES

The repeat-unit SMILES does not encode **chain length (molecular weight)**, **polydispersity**, **tacticity**, **crystallinity** or **processing history**. The target values come from molecular-dynamics simulations, which likely use a fixed simulation protocol for every polymer.